MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, N-CHANNEL, SILICON GATE,

MONOLITHIC 8-BIT MICROPROCESSOR

This amendment forms a part of Military Specification MIL-M-38510/480B, dated 14 April 1983, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.3, Maximum junction temperature: Add Note "1/" after 170°C.

Thermal resistance: Delete "30°C/W" and substitute "(See MIL-M-38510 appendix C). Add "Case X, Y - - - - - 14°C/W 2/".

Bottom of page 1: Add note 1 and 2 as follows:

"1/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening condition per method 5004 of MIL-STD-883.

2/ When a thermal resistance value is included in MIL-M-38510, appendix C, it shall supersede the value stated herein."

* Title: Delete and substitute as printed above.

1.2.3: Following case outline for letter "X" delete "option A".

1.2.3: Delete outline letter "Y" and case outline information that follows it.

* 1.3: Delete "O_{JC}" and substitute "θ_{JC}".

PAGE 3

TABLE I, maximum limits column, V_{OL} test: Delete "0.8" and substitute "0.4".

Minimum limits column, V_{OH} test: Delete "2.0" and substitute "2.4".

Maximum limits column, C_O test: Delete "10" and substitute "15".

Figure no. column starting with t_{cyc} test: Delete "5" and substitute "6". Make this same change on pages 4 and 5.

PAGE 5

TABLE I, maximum limits column for device type 01; t_{PHL9} test: Delete "110" and substitute "120"; t_{PHL13} test, delete "115" and substitute "110".

PAGE 6

Delete note 1/ and substitute: "t_{PLH2} or $t_{PHL2} = t_{PWH1} + t_f - a$".

Device note 10/ and substitute: "$t_{PHL13} = 2\ (t_{cyc}) + t_{PWH1} + t_r - k$".

In constant table: Delete "20, 30, 20" values for H, all device types and substitute "40, 50, 40" respectively. Make these same changes on page 26 notes 8/, 17/ and constant H respectively.

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PAGE 7

* 3.7: Delete and substitute "Marking shall be in accordance with MIL-M-38510."

* 4.2.a.1: After D add "or E" as deleted in Amendment 1.

* TABLE II, delete "Interim" and substitute "Initial".

* TABLE II, class B, for group C and D end point electrical parameters, delete and substitute as follows:

"

Group C end-point electrical parameter (method 5005)	1, 7, 8
Group D end-point electrical parameter (method 5005)	1, 7, 8

"

* TABLE II, class B: For final electrical parameter row, delete "9**" and substitute "9".

PAGE 8

* 4.2.b: Delete "Interim" two places and substitute "Initial".

4.4.1.b, first sentence: Following the word process add "or package".

* 4.4.2, delete and substitute:

"Group B inspection. Group B inspection shall consist of the test subgroups and LTPD values shown in table II of method 5005 of MIL-STD-883 and as follows:

a. A special subgroup shall be added to the group B inspection requirements, using an LTPD of 15 for class B.

b. This test is required only for initial qualification or after redesign. This subgroup shall consist of a high voltage test of the input protection circuits, V_{ZAP} (see 4.5.2)."

PAGE 9

* 4.4.3b (3): Delete "appendix B of MIL-M-38510 and".

4.5.2, first sentence: Delete "$\overline{BUSAK}$" and substitute "$\overline{BUSRQ}$".

* 4.4.3.b.1: After D add "or E" as deleted in Amendment 1.

6.2, delete and substitute as follows:

"6.2 Test vector table. Table V test vectors which forms a part of this detail specification, is not printed herein due to its extreme length and complexity. Table V is a software control program of the sequence of vectors needed for driving the devices represented in this detail sheet as part of the inspection required by table II herein. Table V is only generated when a qualified source exists for this specification. When a qualified source exists, a single hard copy printout of table V is maintained for reference purposes only by the Defense Electronic Supply Center (DESC). DESC-ECS may be consulted for information on obtaining a copy."

PAGE 11

FIGURE 1 title: Following "case outline X" delete "Y".

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PAGE 14

FIGURE 4: Delete in its entirety the table called modes for high voltage test. Add new note as follows "All pins grounded except DUT pin under test".

PAGE 15

NOTE 4.a: Following V_{BIAS}, add "1".

NOTE 4.b: Following the first V_{BIAS}, add "1" following the second V_{BIAS}, add "2".

PAGE 16

TABLE III, case identifier: Delete "Y" make this same change for all pages of table III.

PAGE 21

TABLE III, tests 134-138, 140-147, 150-155, 159-171; maximum limits column: Delete "10" and substitute "15".

PAGE 24

TABLE III, figure number column: Delete "5" and substitute "6" make this same change on page 25.

PAGE 25

TABLE III, test 192 maximum limit column, device type 01: Delete "110" and substitute "120".

Tests 199 and 200, terminal(s) column: Delete "$\overline{MI}$" and substitute "$\overline{MI}$".

Test 199, maximum limits column for device type 01: Delete "115" and substitute "110".

Test 207 terminal(s) column: Delete "$\overline{MERQ}$" and substitute "$\overline{MREQ}$".

PAGE 26

NOTE 18, following 11, add "Tests"; following table, delete "IV" and substitute "IVA".

NOTE 6, following table IV: Add a period and delete remainder of sentence.

Footnote 18/, delete and substitute:

"18/ These ac limits for subgroup 12 are clock dependent (clock rising or falling; high or low; or cycle time. The equations represented by notes 8-17 herein shall be used to calculate limits from F_{MAX} input conditions A and B and F_{MIN} input conditions C and D of table IV.

PAGE 30

TABLE IV heading: Delete "IV" and substitute "IVA".

Test symbol t_{PLH13}, t_{PHL13} reference 20, 19 terminal(s) column: Delete "$\overline{MI}$" and substitute "$\overline{MI}$".

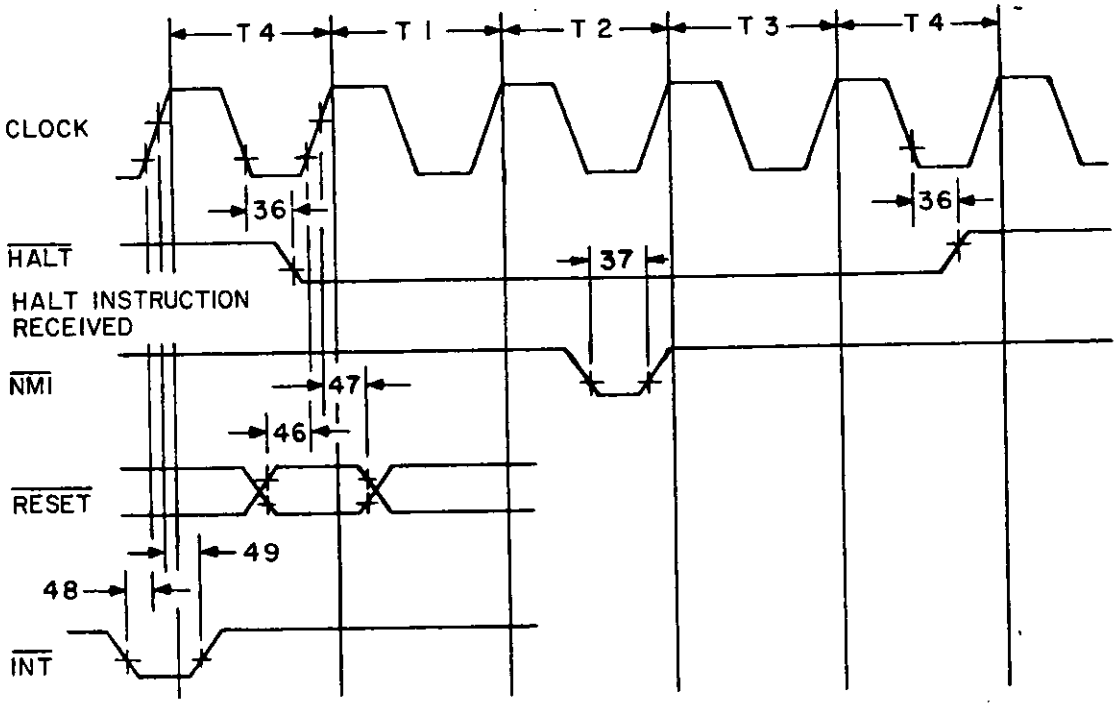
PAGE 37

Figure 6: Delete "I/O read" and substitute "I/O write".

PAGE 38

FIGURE 6: Delete "I/O write" and substitute "I/O read".

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NOTE: SIGNALS ARE NOT NECESSARILY RELATED

FIGURE 6. Timing diagram - Continued.

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PAGE 42

Delete and substitute new page as printed on page 4 of this amendment.

NOTE: The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
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